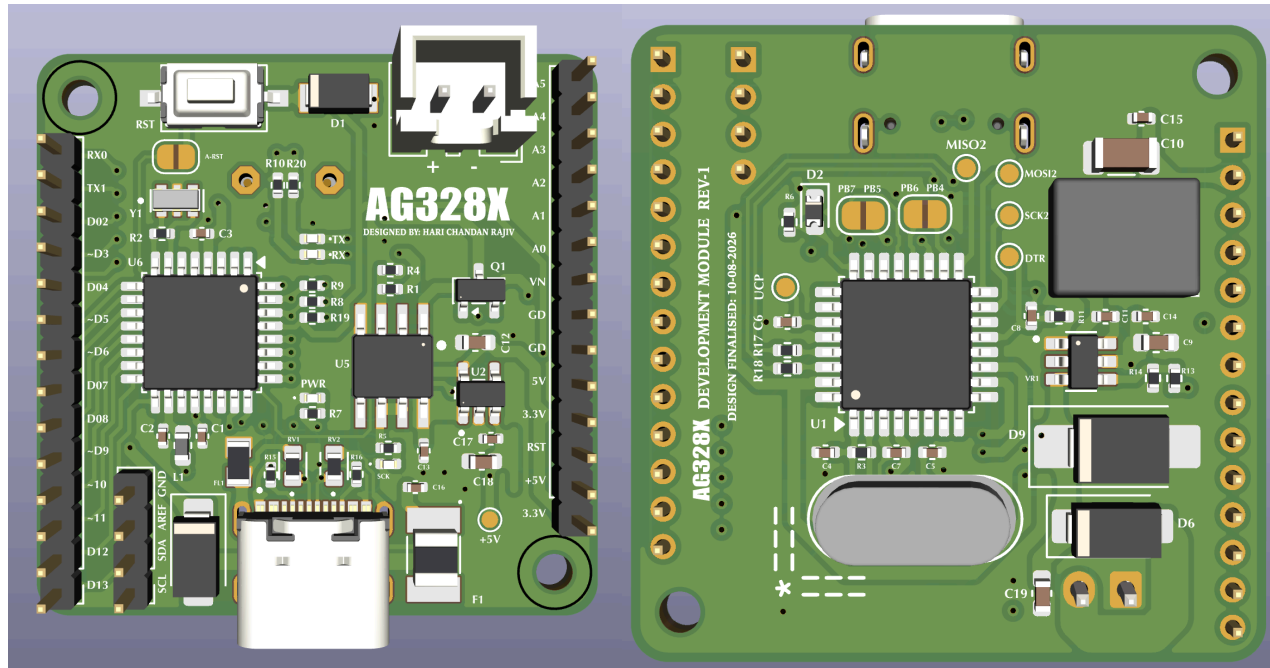


AG328X REV-1 Description



1.1 Introduction

AG328X REV-1 is a compact, redesigned implementation of the Arduino UNO platform, developed with the objective of improving the original UNO architecture in terms of form factor, power architecture, USB connectivity, protection, serviceability, and PCB integration.

The design retains the core functionality based on the ATmega328P, while keeping a dedicated ATmega16U2 USB-to-serial interface and a redesigned power-management architecture. Rather than directly reproducing the conventional reference design, AG328X restructures several functional sections to achieve a more compact and integrated hardware platform.

The complete PCB is designed with an overall dimension of 34 mm × 34 mm, a 1.6 mm finished board thickness, and a 4-layer PCB stack-up, which is **68.44%** smaller in size compared to the original UNO board. The four-layer implementation provides additional routing resources and dedicated copper areas for power and ground distribution, enabling the compact board dimensions without compromising the electrical organization of the design.

1.2 Design Objective

The primary objective of AG328X REV-1 is to develop a smaller and more technically refined UNO-compatible hardware platform while maintaining the essential functionality. The redesign focuses on the following engineering objectives:

- Reduction of the overall PCB footprint.
- Integration of the complete ATmega328P-based controller architecture onto a compact board.
- Replacement of the conventional USB-B connector arrangement with USB Type-C.
- Improvement of the input power and regulation architecture.
- Improvement on maximum input voltage VIN capability.
- Replacement of DC jack with a 2 pin JST connector for easy power connection.
- Controlled generation of the 3.3 V rail, including VIN/USBVCC source selection.
- Incorporation of input protection and power filtering.
- Improved PCB routing capability through a 4-layer architecture.
- Maintaining access to the standard Arduino I/O functionality through board connectors.

1.3 System Architecture

The AG328X REV-1 hardware is organized around two primary microcontrollers:

1. ATmega328P — Main application microcontroller responsible for the Arduino-compatible processing and I/O functionality.
2. ATmega16U2 — Dedicated USB interface controller responsible for USB-to-UART communication with the ATmega328P.

The schematic is divided into functional blocks covering the USB interface, power management, microcontroller circuitry, indication circuitry, and external I/O interfaces. The power architecture is independently organized around the incoming VIN and USB VCC sources. The design includes a dedicated 5 V regulator section for generating the primary regulated supply and a separate 3.3 V supply section with source-selection circuitry. Input protection, filtering, bulk capacitance, and local decoupling are incorporated around the power stages to improve supply integrity.

The USB section uses a USB Type-C connector together with the required CC configuration and USB signal routing. A ferrite-bead-based filtering arrangement is also incorporated into the USB-related power/ground path.

2. Technical Specifications

2.1 Electrical and System Specifications

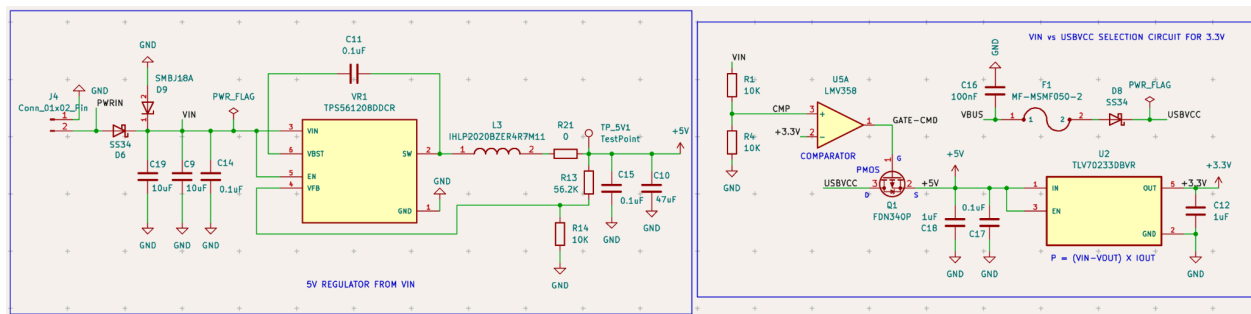
Parameter	AG328X REV-1 Specification	Design Basis
Main Controller	ATmega328P-AU	32 KB Flash, 2 KB SRAM, 1 KB EEPROM
USB Controller	ATmega16U2	Dedicated USB-to-UART interface
Main Logic Voltage	5 V	Regulated supply
Auxiliary Logic Voltage	3.3 V	Dedicated LDO
VIN Input	17 V maximum	Regulator Input capability
5 V Regulator Input Capability	4.5–17 V	TPS561208-class buck regulator
3.3 V LDO Input	5.5 V maximum	TLV70233
USB Interface	USB 2.0	ATmega16U2

USB Connector	USB Type-C	For easy power and coding accessibility
PCB Dimensions	34 mm × 34 mm	Mechanical design
Reverse Polarity Protection	FDN340P	PMOS transistor based reverse polarity and dual input protection
PCB Thickness	1.6 mm	PCB specification
PCB Layer Count	4 layers	Routing and ground/power architecture

The ATmega328P supports a 1.8–5.5 V supply range, while the ATmega16U2 supports 2.7–5.5 V. Both therefore operate within their specified supply ranges on the 5 V AG328X logic rail. ([Microchip](#))

2.2 VIN Capability and Input Protection

The VIN power path is one of the major improvements in AG328X REV-1. The input passes through a resettable PPTC fuse, reverse-polarity protection diode, and transient-voltage suppression device before reaching the regulator stage.



VIN Rating Analysis

The selected 5 V switching regulator provides an input operating range extending to 17 V. For the TPS561201, for example, the specified input range is 4.5–17 V, with a 1 A output capability. ([Texas Instruments](#))

However, the overall AG328X VIN rating is determined by the lowest-rated component in the VIN path, not by the regulator alone.

The selected Bourns MF-MSMF050-2 has:

- Hold current: 500 mA
- Trip current: 1 A
- Maximum voltage: 15 VDC
- Maximum current: 100 A
- Maximum resistance: approximately 1 Ω ([Mouser Electronics](#))

Therefore: The present AG328X REV-1 VIN capability should be 15 V maximum, not 17 V.

The 17 V capability of the buck regulator provides voltage headroom within the regulator itself, but the 15 V PPTC rating limits the complete input path.

Input Protection Devices

The VIN protection network includes the following devices:

Component	Selected Part	Rating / Function
F1	MF-MSMF050-2	500 mA hold, 1 A trip, 15 V maximum
D6	SS34	Schottky reverse-polarity protection
D9	SMBJ18A	18 V TVS protection
C19/C9/C14	Input capacitors	VIN filtering and transient energy storage

The SMBJ18A provides an 18 V working standoff voltage, 20 V minimum breakdown voltage and approximately 29.2 V maximum clamping voltage, with a 600 W peak pulse capability for the specified pulse condition. ([Diodes Incorporated](#))

This provides appropriate transient protection around the intended VIN region, but it does not change the continuous operating-voltage limit imposed by the PPTC fuse.

2.3 5 V Power Supply

AG328X REV-1 uses a dedicated synchronous buck regulator to generate the primary 5 V rail from VIN.

The selected regulator is a TPS56x-series 17 V-input synchronous step-down converter. The relevant device family provides:

- Input voltage: 4.5–17 V
- Output voltage range: approximately 0.76–7 V
- Synchronous rectification
- Integrated switching MOSFETs
- Cycle-by-cycle overcurrent protection
- Hiccup-mode overcurrent protection
- Thermal shutdown
- Undervoltage protection
- Approximately 580 kHz switching frequency for the TPS56120x family. ([Texas Instruments](#))

The regulator is configured for a fixed 5 V output, which becomes the primary supply rail for the ATmega328P, ATmega16U2 and associated 5 V circuitry.

System Implication

The regulator selection provides considerably more input-voltage capability than the conventional 5 V logic rail requires:

VIN → Protection → 5 V Buck Converter → +5 V Logic Rail

The important distinction is:

- Regulator capability: up to 17 V
- Present AG328X system VIN rating: 15 V
- Reason: MF-MSMF050-2 maximum voltage rating

This distinction is maintained throughout this document to prevent confusing individual component ratings with the system rating.

2.4 3.3 V Power Supply

The 3.3 V rail is generated using the TLV70233 low-dropout regulator.

The TLV70233 is a fixed 3.3 V LDO with:

- Output voltage: 3.3 V
- Output current capability: 300 mA
- Maximum input voltage: 5.5 V
- Low quiescent current
- Low-noise operation
- Thermal and current protection. ([Texas Instruments](#))

In AG328X, the regulator is supplied from the 5 V domain, rather than directly from VIN.

Therefore, the 3.3 V regulator does not impose a 5.5 V limit on the AG328X VIN input.

The power architecture can consequently be represented as:

VIN → 5 V Buck → 5 V Rail → 3.3 V LDO → 3.3 V Rail

This allows the board to accept the higher VIN voltage permitted by the upstream power architecture while keeping the 3.3 V regulator safely within its input specification.

2.5 3.3 V Supply-Source Selection

AG328X REV-1 incorporates an active supply-selection circuit around an LMV358 comparator and FDN340P P-channel MOSFET to control the 3.3 V supply source.

The LMV358 operates from 2.7–5.5 V and provides rail-to-rail output capability, making it suitable for the 5 V-powered comparison/control section used in the design. ([Texas Instruments](#))

The selected FDN340P is a logic-level P-channel MOSFET with:

- Drain-source voltage: 20 V
- Continuous drain current: approximately 2 A
- $R_{DS(on)}$: approximately 70 m Ω at $V_{GS} = -4.5$ V
- Maximum gate-source voltage: ± 8 V
- Power dissipation: approximately 500 mW
- SOT-23 package. ([DigiKey](#))

This provides adequate voltage margin for the 5 V-domain source-selection function.

2.6 5 V Regulator Architecture and Power-Loss Improvement

A significant change in AG328X REV-1 is the replacement of the conventional 5 V linear/LDO regulation approach with a switching buck regulator.

This change was made primarily to reduce power dissipation and improve power-conversion efficiency, particularly when AG328X is operated from a higher VIN voltage.

Linear Regulator vs. Switching Regulator

With a conventional linear regulator, the excess input voltage is dissipated directly as heat. The regulator power loss is approximately:

$$P_{LOSS(LDO)} = (V_{IN} - V_{OUT}) \times I_{OUT}$$

For the AG328X 5 V rail:

$$P_{LOSS(LDO)} = (V_{IN} - 5 \text{ V}) \times I_{OUT}$$

As V_{IN} increases, the voltage dropped across the regulator increases, causing a corresponding increase in heat generation.

In contrast, the switching buck regulator used in AG328X transfers energy from the input to the output through high-frequency switching. Its power loss is primarily determined by the converter's efficiency:

$$P_{LOSS(BUCK)} = P_{OUT} \times (1/\eta - 1)$$

where:

- P_{OUT} = output power
- η = converter efficiency

Therefore, the switching architecture avoids dissipating the entire VIN-to-5 V voltage difference as heat.

Example Power-Loss Comparison

Consider a representative operating condition of:

- $V_{IN} = 15 \text{ V}$
- $V_{OUT} = 5 \text{ V}$
- $I_{OUT} = 500 \text{ mA}$

The 5 V output power is:

$$P_{OUT} = 5 \text{ V} \times 0.5 \text{ A} = 2.5 \text{ W}$$

a. Conventional LDO

For a linear regulator: $P_{LOSS} = (15 - 5) \times 0.5 = 5 \text{ W}$

The regulator would therefore need to dissipate approximately 5 W as heat while delivering only 2.5 W to the 5 V load. The theoretical efficiency would be:

$$\eta_{LDO} = 5/15 \approx 33.3\%$$

This creates a significant thermal-management requirement and makes operation from higher VIN voltages inefficient.

b. Switching Buck Regulator

Assuming a representative converter efficiency of 90%:

$$P_{IN} = 2.5 \text{ W} / 0.90 \approx 2.78 \text{ W}$$

Therefore: $P_{LOSS} \approx 2.78 - 2.5 = 0.28 \text{ W}$

Parameter	5 V LDO	5 V Switching Regulator
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VIN	15 V	15 V
VOUT	5 V	5 V
Load Current	500 mA	500 mA
Output Power	2.5 W	2.5 W
Approx. Efficiency	33.3%	~90%*
Approx. Regulator Loss	5.0 W	0.28 W*
Heat Generation	High	Significantly lower

Actual switching-regulator efficiency depends on VIN, load current, switching frequency, inductor, PCB layout, temperature, and other operating conditions. The 90% value is an illustrative engineering example and should not be treated as the guaranteed AG328X efficiency.

Effect on AG328X

The change to a switching 5 V regulator provides several practical advantages:

1. **Lower power dissipation**

The voltage difference between VIN and 5 V is no longer converted predominantly into heat.

2. **Improved efficiency at higher VIN**

The advantage becomes increasingly significant as VIN increases.

3. **Reduced thermal stress**

Lower regulator dissipation reduces junction-temperature rise and thermal stress on the PCB.

4. **Improved usable VIN range**

The selected switching regulator supports input voltages up to **17 V**, while the present AG328X system-level VIN rating is limited to **15 V by the MF-MSMF050-2 input protection fuse**.

5. **Better suitability for continuous operation**

Lower dissipation is particularly advantageous when the board operates continuously or is enclosed with limited airflow.

6. **Higher practical power capability**

Because less input power is lost as heat, a larger portion of the available input power can be delivered to the 5 V rail.

Engineering Significance

The regulator change is therefore not simply a component substitution. It represents a **power-architecture improvement** in AG328X REV-1.

Design conclusion: The adoption of a switching regulator for the 5 V rail significantly reduces the power that would otherwise be dissipated as heat by a linear regulator, making the AG328X power architecture more efficient and thermally suitable for operation from higher VIN voltages.

2.7 Current Consumption and Power-Architecture Comparison

A further objective of AG328X REV-1 is to reduce unnecessary power consumption and, more importantly, reduce the power lost within the board's voltage-regulation circuitry.

Arduino UNO R3 Reference

The Arduino UNO R3 is specified for a 5 V operating voltage and uses a linear regulation approach when powered through VIN. Arduino's current power-supply guidance indicates that a basic Arduino board typically requires approximately **50 mA**, excluding external loads connected to the board. ([Arduino Help Center](#))

The actual input power when using VIN depends strongly on the applied voltage because a linear regulator dissipates the voltage difference as heat.

For example, at:

- VIN = **12 V**
- 5 V rail current = **50 mA**

the approximate regulator dissipation is: $P_{LOSS} = (12 - 5) \times 0.05 = 0.35 \text{ W}$

The input power associated with the 50 mA load is approximately: $P_{IN} = 12 \times 0.05 = 0.60 \text{ W}$

while the useful 5 V power is: $P_{OUT} = 5 \times 0.05 = 0.25 \text{ W}$

This illustrates why the UNO's linear-regulator architecture becomes increasingly inefficient as V_{IN} increases.

AG328X REV-1

AG328X replaces the conventional 5 V linear-regulation approach with a **switching buck regulator**. Consequently, the input current is no longer approximately equal to the 5 V load current when operating from V_{IN} .

For a buck converter:

$$I_{IN} \approx (V_{OUT} \times I_{OUT}) / (V_{IN} \times \eta)$$

where η is the converter efficiency.

For example, assuming:

- $V_{IN} = 12 \text{ V}$
- 5 V load current = **50 mA**
- Converter efficiency = **90%**

the approximate V_{IN} current becomes:

$$I_{IN} \approx (5 \times 0.05) / (12 \times 0.90) \approx 23 \text{ mA}$$

The corresponding input power is approximately:

$$P_{IN} \approx 12 \times 23 \text{ mA} \approx 0.28 \text{ W}$$

Therefore, compared with a linear regulator, the switching architecture substantially reduces the power drawn from V_{IN} for the same 5 V load and dramatically reduces regulator heat dissipation.

Preliminary Comparison

Parameter	Arduino UNO R3	AG328X REV-1
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Main MCU	ATmega328P	ATmega328P
USB MCU	ATmega16U2	ATmega16U2
Logic Voltage	5 V	5 V
5 V Regulation	Linear	Switching Buck
Typical board current*	~50 mA	To be measured
VIN example	12 V	12 V
5 V load used for example	50 mA	50 mA
Approx. VIN current at 12 V	~50 mA	~23 mA*
Approx. regulator loss	~350 mW	~28 mW*
Regulation efficiency	~42% theoretical	~90% assumed*

The AG328X value is an engineering calculation based on an assumed 50 mA 5 V load and 90% converter efficiency. It is **not a measured AG328X specification**.

Engineering Conclusion

The principal improvement is therefore not necessarily that **AG328X consumes fewer milliamps at every operating condition**. The more significant improvement is that AG328X requires **less input power and generates substantially less heat when converting higher VIN voltages to the 5 V logic rail**.

For the same 5 V load:

Arduino UNO linear regulation:

Higher VIN → approximately same input current → higher power loss

AG328X switching regulation:

Higher VIN → lower input current → substantially lower power loss

This makes the AG328X architecture more suitable for compact applications where **thermal dissipation, continuous operation, and power efficiency** are important design considerations.